GATE DA 2027

Linear Algebra

Vector spaces, matrices, systems, eigenvalues, quadratic forms, LU and SVD with the stories of Ramu and Somu

The two friends are back.
Ramu asks "but why?" about everything. **Somu** explains with small everyday examples. Linear algebra is the maths of *many numbers at once*, so most of our stories are about bags of fruit, shops, and a small town's roads.

Syllabus covered (GATE DA, Section 2): vector spaces, matrix types, quadratic forms, linear systems, eigenvalues and eigenvectors, rank and nullity, LU decomposition and SVD.

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1 Vectors and Vector Spaces

Ramu & Somu: Fruit shopping lists

Ramu goes to the market and buys **2 apples and 3 bananas**. Somu writes this as a vector $\begin{pmatrix}2\\3\end{pmatrix}$. Another day Ramu buys $\begin{pmatrix}1\\0\end{pmatrix}$ (one apple) and $\begin{pmatrix}0\\1\end{pmatrix}$ (one banana). Any shopping list can be built from these two: $\begin{pmatrix}2\\3\end{pmatrix} = 2\begin{pmatrix}1\\0\end{pmatrix} + 3\begin{pmatrix}0\\1\end{pmatrix}$. This is a **linear combination**, and the two basic lists form a **basis** for the "fruit space" $\mathbb{R}^2$.

But if Ramu is given the lists $\begin{pmatrix}1\\2\end{pmatrix}$ and $\begin{pmatrix}2\\4\end{pmatrix}$, the second is just double the first. No matter how he combines them, he can only produce lists along one line. They are **linearly dependent**: one of them is wasted.

1.1 Vector space and subspace

Definitions

A **vector space** over $\mathbb{R}$ is a set closed under addition and scalar multiplication (with the usual axioms: zero vector, additive inverse, associativity, distributivity, etc.).

A **subspace** $W \subseteq V$ must satisfy: (i) $0 \in W$; (ii) $u, v \in W \Rightarrow u + v \in W$; (iii) $u \in W,\ c \in \mathbb{R} \Rightarrow cu \in W$.

Examples of subspaces of $\mathbb{R}^n$: the zero set $\{0\}$, lines and planes through the origin, the solution set of $A\mathbf{x} = \mathbf{0}$, the column space of any matrix. **Non-examples:** a line not through 0; the set $\{x : x \geq 0\}$ (not closed under negatives).

1.2 Span, independence, basis, dimension

Formulas

- Linear combination: $c_1v_1 + c_2v_2 + \cdots + c_kv_k$
- $\text{span}\{v_1, \ldots, v_k\}$ = set of all linear combinations (a subspace)
- Linearly independent: $c_1v_1 + \cdots + c_kv_k = 0 \ \Rightarrow\ c_1 = \cdots = c_k = 0$
- Basis: linearly independent *and* spans V
- $\dim V$ = number of vectors in any basis (all bases have the same size)
- $\dim \mathbb{R}^n = n$, $\dim(\text{ space of } m \times n \text{ matrices }) = mn$, $\dim(\text{ polynomials of degree} \leq d) = d + 1$

Idea in plain words

Independence means *no vector is redundant*. Practical test: put the vectors as columns of a matrix A. They are independent exactly when $A\mathbf{c} = \mathbf{0}$ has only the solution $\mathbf{c} = \mathbf{0}$, i.e. when $\text{rank}(A)$ equals the number of vectors. In $\mathbb{R}^n$, any set of more than n vectors is dependent; any n independent vectors form a basis.

Independent or not?

$v_1 = (1, 2, 3)$, $v_2 = (2, 4, 6)$, $v_3 = (1, 0, 1)$. Since $v_2 = 2v_1$, the set is dependent. $\text{span}\{v_1, v_2, v_3\} = \text{span}\{v_1, v_3\}$ has dimension 2 (a plane in $\mathbb{R}^3$).

1.3 The four fundamental subspaces

Ramu & Somu: A shop's price list

Somu runs a small stall with a matrix A ($m \times n$): each row is a customer and each column an item. Ramu asks about four natural questions:

- Which totals $\mathbf{b}$ can the stall produce? $\Rightarrow$ the **column space**.
- Which combinations of items give a zero bill? $\Rightarrow$ the **null space**.

Row space and left null space are the same two questions for $A^\top$.

Formulas

For an $m \times n$ matrix A with $\text{rank}(A) = r$:

Subspace	Lives in	Dimension	Meaning
Column space $\mathcal{C}(A)$	$\mathbb{R}^m$	r	all $A\mathbf{x}$
Null space $\mathcal{N}(A)$	$\mathbb{R}^n$	$n - r$	all $\mathbf{x}$ with $A\mathbf{x} = 0$
Row space $\mathcal{C}(A^\top)$	$\mathbb{R}^n$	r	span of rows
Left null space $\mathcal{N}(A^\top)$	$\mathbb{R}^m$	$m - r$	all $\mathbf{y}$ with $A^\top\mathbf{y} = 0$

Orthogonality: row space $\perp$ null space (in $\mathbb{R}^n$); column space $\perp$ left null space (in $\mathbb{R}^m$).

1.4 Inner product, norm, orthogonality, projection

Formulas

$$\langle u, v\rangle = u^\top v = \sum u_i v_i, \qquad \|u\| = \sqrt{u^\top u}, \qquad \cos\theta = \frac{u^\top v}{\|u\|\|v\|}$$

Cauchy–Schwarz: $|u^\top v| \le \|u\|\|v\|$, Triangle: $\|u + v\| \le \|u\| + \|v\|$

$u \perp v \iff u^\top v = 0$ (orthogonal nonzero vectors are independent)

Projection of b onto a : $\text{proj}_a\, b = \dfrac{a^\top b}{a^\top a}\, a, \qquad P = \dfrac{aa^\top}{a^\top a}$

Projection onto $\mathcal{C}(A)$ (A full column rank) : $P = A(A^\top A)^{-1}A^\top$

Least squares: minimise $\|Ax - b\|^2 \Rightarrow$ normal equations $A^\top A\,\hat{x} = A^\top b$

Ramu & Somu: Making directions perpendicular: Gram–Schmidt

Ramu has two directions $v_1 = (1, 1, 0)$ and $v_2 = (1, 0, 1)$ and wants a pair of *perpendicular unit* directions covering the same plane. Somu: keep the first direction, then remove from the second

whatever lies along the first.

$$e_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{2}}(1,1,0), \qquad w_2 = v_2 - (v_2^\top e_1)e_1 = (1,0,1) - \tfrac{1}{2}(1,1,0) = (\tfrac{1}{2}, -\tfrac{1}{2}, 1),$$

$$e_2 = \frac{w_2}{\|w_2\|} = \frac{1}{\sqrt{3/2}}(\tfrac{1}{2}, -\tfrac{1}{2}, 1).$$

Check $e_1^\top e_2 \propto \frac{1}{2} - \frac{1}{2} = 0$. This is the **Gram–Schmidt process**, giving $A = QR$ (Q orthonormal columns, R upper triangular).

Common GATE traps

(1) A subspace must contain 0. (2) Dependent vectors can still span (they just contain redundancy). (3) dim of the solution set of $A\mathbf{x} = \mathbf{b}$ ($\mathbf{b} \neq 0$) is not a vector-space dimension in the same sense: it is a *shifted* subspace (an affine set) with the same "size" $n - r$.

2 Matrix Types and Basic Operations

Ramu & Somu: A town's roads

Somu draws a map of 3 villages and writes a 3×3 matrix where entry (i, j) is 1 if there is a road from village i to j. Ramu notices: if all roads are two-way, the matrix equals its own mirror image across the diagonal. Such a matrix is **symmetric**. Different shapes of matrix tell us different stories, and each has a name.

2.1 Catalogue of special matrices

Type	Definition	Key facts
Diagonal	$a_{ij} = 0$ for $i \neq j$	eigenvalues = diagonal entries; det $= \prod d_i$
Identity I	diagonal with 1's	$AI = IA = A$
Upper / lower triangular	zeros below / above diagonal	eigenvalues = diagonal entries; det $= \prod$ diagonal
Symmetric	$A^\top = A$	real eigenvalues; orthogonal eigenvectors; diagonalisable
Skew-symmetric	$A^\top = -A$	diagonal $= 0$; eigenvalues purely imaginary or 0; odd order $\Rightarrow$ det $= 0$
Orthogonal	$Q^\top Q = QQ^\top = I$	$Q^{-1} = Q^\top$; det $= \pm 1$; $\lvert\lambda\rvert = 1$; preserves lengths
Idempotent	$A^2 = A$	eigenvalues 0 or 1; projections
Nilpotent	$A^k = 0$ for some k	all eigenvalues 0
Involutory	$A^2 = I$	eigenvalues ± 1
Singular	$\det A = 0$	not invertible; has eigenvalue 0; rank $< n$
Invertible (non-singular)	$\det A \neq 0$	rank $= n$; unique solution to $A\mathbf{x} = \mathbf{b}$
Positive definite	symmetric, $x^\top Ax > 0\ \forall x \neq 0$	all eigenvalues > 0 (see Section 6)
Hermitian (complex)	$A^* = A$	real eigenvalues
Unitary (complex)	$U^*U = I$	complex analogue of orthogonal

2.2 Rules of matrix arithmetic

Formulas

$$(AB)^\top = B^\top A^\top, \qquad (AB)^{-1} = B^{-1}A^{-1}, \qquad (A^\top)^{-1} = (A^{-1})^\top$$

$AB \neq BA$ in general; $\quad (A+B)^2 \neq A^2 + 2AB + B^2$ unless $AB = BA$

$$\operatorname{tr}(A) = \sum a_{ii}, \qquad \operatorname{tr}(AB) = \operatorname{tr}(BA), \qquad \operatorname{tr}(A+B) = \operatorname{tr} A + \operatorname{tr} B$$

Any square $A = \underbrace{\tfrac{1}{2}(A + A^\top)}_{\text{symmetric}} + \underbrace{\tfrac{1}{2}(A - A^\top)}_{\text{skew-symmetric}}$

2×2 inverse: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \dfrac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

$$A^{-1} = \frac{\operatorname{adj}(A)}{\det A}, \qquad \operatorname{adj}(A) = \text{transpose of the cofactor matrix}$$

2.3 Determinant

Formulas

- $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$
- $\det(AB) = \det A \det B, \qquad \det(A^\top) = \det A, \qquad \det(A^{-1}) = \frac{1}{\det A}$
- $\det(cA) = c^n \det A \quad (n \times n), \qquad \det(A^k) = (\det A)^k$
- Row swap: det $\to -\det$; scale a row by c: det $\to c\det$; add a multiple of one row to another: det unchanged.
- Two equal (or proportional) rows or a zero row $\Rightarrow \det = 0$
- $\det A = \prod_i \lambda_i, \qquad \operatorname{tr} A = \sum_i \lambda_i \qquad$ (eigenvalues with multiplicity)

Worked example

$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$: $\det A = 4 - 6 = -2$, $\det(2A) = 2^2(-2) = -8$, $A^{-1} = \frac{1}{-2}\begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix}$.

3 Linear Systems $A\mathbf{x} = \mathbf{b}$

Ramu & Somu: Ramu's three shops

Ramu buys 3 kinds of fruit from a stall but forgets the prices. He remembers three bills:

$$x + y + z = 6, \qquad 2y + 5z = -4, \qquad 2x + 5y - z = 27.$$

(Here the numbers are just for practice.) Somu solves it by **Gaussian elimination**: use the first equation to remove x from the others, then use the next to remove y, then back-substitute. Result: $x = 5,\ y = 3,\ z = -2$. Check: $5 + 3 - 2 = 6$, $6 - 10 = -4$, $10 + 15 + 2 = 27$.

3.1 Matrix form and elimination

Formulas

- $A\mathbf{x} = \mathbf{b}$, $A \in \mathbb{R}^{m \times n}$ (m equations , n unknowns), augmented matrix $[A \mid \mathbf{b}]$
- Row operations: swap rows; scale a row; add a multiple of one row to another (do not change the solution set)
- Gauss elimination $\rightarrow$ row echelon form Gauss–Jordan $\rightarrow$ reduced row echelon form (RREF)
- Pivot columns $\leftrightarrow$ basic variables; non-pivot columns $\leftrightarrow$ free variables

3.2 When does a solution exist? (The master table)

Rouché–Capelli theorem

Let $r = \text{rank}(A)$, $r_{aug} = \text{rank}([A \mid \mathbf{b}])$, $n =$ number of unknowns.

Condition	Result	Free variables
$r < r_{aug}$	**no solution** (inconsistent)	—
$r = r_{aug} = n$	**unique solution**	0
$r = r_{aug} < n$	**infinitely many solutions**	$n - r$

Always $r \le r_{aug} \le r + 1$.

3.3 Homogeneous systems $A\mathbf{x} = \mathbf{0}$

Formulas

- Always consistent (the zero solution). It has a **non-trivial** solution $\iff r < n$.
- If $m < n$ (fewer equations than unknowns), there are always non-trivial solutions.
- For square A: non-trivial solution $\iff \det A = 0$.
- Solution set $= \mathcal{N}(A)$, a subspace of dimension $n - r$.
- General solution of $A\mathbf{x} = \mathbf{b}$: $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$ (one particular solution plus anything in the null space).

Cramer's rule (square, $\det A \neq 0$): $x_i = \dfrac{\det A_i}{\det A}$, where A_i is A with column i replaced by $\mathbf{b}$.

Inconsistent vs infinite

$x + y = 2,\ x + y = 3$: $r = 1$, $r_{aug} = 2$, **no solution** (parallel lines).
$x + y = 2,\ 2x + 2y = 4$: $r = r_{aug} = 1 < 2$, so **infinitely many** (1 free variable): $(2 - t,\ t)$.
Three equations, five unknowns, rank $= 3$: $5 - 3 = 2$ free variables, infinitely many solutions.

Common GATE traps

(1) Comparing only rank(A) with n is not enough; *always compare with* rank$[A|\mathbf{b}]$. (2) A square system with $\det A = 0$ can have either no solution or infinitely many. (3) "Free variables $= n -$ rank" counts unknowns, not equations.

4 Rank and Nullity

Ramu & Somu: Duplicate shopkeepers

Somu has a table of 4 shopkeepers' price lists (4 rows). But two of the shopkeepers simply copy each other, and the fourth always quotes exactly the sum of the other two. Ramu asks, "How many *truly different* shopkeepers are there?" The answer is the **rank**: the number of independent rows (equivalently, independent columns). A rank-deficient matrix hides redundancy.

Formulas

- $\operatorname{rank}(A) =$ number of pivots $= \dim \mathcal{C}(A) = \dim($ row space $) =$ max number of independent rows (or columns)
- $\operatorname{nullity}(A) = \dim \mathcal{N}(A) = n - \operatorname{rank}(A)$ (**Rank–Nullity theorem**, $A \in \mathbb{R}^{m\times n}$)
- $\operatorname{rank}(A) \le \min(m, n)$, $\operatorname{rank}(A) = \operatorname{rank}(A^\top)$
- $\operatorname{rank}(AB) \le \min(\operatorname{rank} A, \operatorname{rank} B)$, $\operatorname{rank}(A + B) \le \operatorname{rank} A + \operatorname{rank} B$
- $\operatorname{rank}(A^\top A) = \operatorname{rank}(AA^\top) = \operatorname{rank}(A)$
- Sylvester: $\operatorname{rank} A + \operatorname{rank} B - n \le \operatorname{rank}(AB)$ $(A : m \times n,\ B : n \times p)$
- Square A : invertible $\iff$ rank $= n \iff \det \neq 0 \iff$ columns independent $\iff 0$ is not an eigenvalue
- Rank-1 matrix: $A = uv^\top$ (outer product) $\Rightarrow$ rank $= 1$, only nonzero eigenvalue $v^\top u$

Worked example

$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix}$. Row 2 is $2\times$ row 1, so $\operatorname{rank} A = 2$, nullity $= 3 - 2 = 1$, $\det A = 0$. Null space: $x + 2y + 3z = 0$ and $x + z = 0 \Rightarrow x = -z,\ y = -z$, i.e. spanned by $(-1, -1, 1)$. Check: $-1 - 2 + 3 = 0$, $-1 + 0 + 1 = 0$.
For a 4×6 matrix with rank $= 3$: nullity $= 6 - 3 = 3$; left null space has dimension $4 - 3 = 1$.

Common GATE traps

Rank is found by row-reducing but **row operations do not change the row space or rank**, while they *do* change the column space (its dimension stays the same). $\operatorname{rank}(A + B)$ is only bounded, not equal to the sum.

5 Eigenvalues and Eigenvectors

Ramu & Somu: A machine that only stretches

Ramu has a machine that transforms points of the plane: $\mathbf{x} \mapsto A\mathbf{x}$. For most arrows the machine *turns and stretches* them. But Somu finds special arrows that the machine only *stretches* without turning: $A\mathbf{v} = \lambda\mathbf{v}$. These are **eigenvectors**, and the stretch factors λ are **eigenvalues**. ("Eigen" means "own" in German: the matrix's own special directions.)

5.1 Definition and how to compute

Formulas

$A\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{v} \neq 0 \iff (A - \lambda I)\mathbf{v} = 0$ has a nonzero solution $\iff \det(A - \lambda I) = 0$

Characteristic polynomial: $p(\lambda) = \det(\lambda I - A) = \lambda^n - (\operatorname{tr} A)\lambda^{n-1} + \cdots + (-1)^n \det A$

For 2×2 : $\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0$

Eigenvectors for λ : non-zero vectors in $\mathcal{N}(A - \lambda I)$ (the eigenspace)

5.2 Properties you must know

Formulas

$\sum \lambda_i = \operatorname{tr} A, \qquad \prod \lambda_i = \det A$

Eigenvalues of A^k : λ^k; $\quad A^{-1}$: $1/\lambda$; $\quad A + cI$: $\lambda + c$; $\quad cA$: $c\lambda$; $\quad p(A)$: $p(\lambda)$

A and $A^\top$ have the same eigenvalues (not the same eigenvectors)

AB and BA have the same nonzero eigenvalues

Triangular/diagonal: eigenvalues are the diagonal entries

Eigenvectors for *distinct* eigenvalues are linearly independent

Symmetric real A : real eigenvalues; eigenvectors for distinct eigenvalues are orthogonal

Singular $\iff$ 0 is an eigenvalue; $\quad$ Idempotent: $\lambda \in \{0, 1\}$; $\quad$ Orthogonal: $|\lambda| = 1$

Cayley–Hamilton: $p(A) = 0$ (every matrix satisfies its own characteristic equation)

5.3 Multiplicities and diagonalisation

Formulas

- Algebraic multiplicity (AM) = multiplicity of λ as a root of $p(\lambda)$
- Geometric multiplicity (GM) $= \dim \mathcal{N}(A - \lambda I), \qquad 1 \leq \text{GM} \leq \text{AM}$
- A is **diagonalisable** $\iff$ GM = AM for every eigenvalue $\iff$ A has n independent eigenvectors
- Then $A = PDP^{-1}$, $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, $P = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_n]$, $\quad A^k = PD^kP^{-1}$
- Symmetric $A = Q\Lambda Q^\top$ (spectral theorem, Q orthogonal)

Ramu & Somu: Powers of a machine

Ramu applies the machine $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ to a point 10 times. Multiplying matrices 10 times is painful, so Somu finds the eigenvalues: $\operatorname{tr} A = 7$, $\det A = 10$, so $\lambda^2 - 7\lambda + 10 = 0 \Rightarrow \lambda = 5, 2$.

For $\lambda = 5$: $(A - 5I) = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \Rightarrow \mathbf{v}_1 = (1, 1)$.

For $\lambda = 2$: $(A - 2I) = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \Rightarrow \mathbf{v}_2 = (1, -2)$.

Check: $A(1, -2) = (4 - 2,\ 2 - 6) = (2, -4) = 2(1, -2)$. ✓

So $A = PDP^{-1}$ with $P = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}$, $D = \operatorname{diag}(5, 2)$, and $A^{10} = P \operatorname{diag}(5^{10}, 2^{10}) P^{-1}$: just raise two numbers to the 10th power. In the direction $\mathbf{v}_1$ everything grows by 5 each step; in $\mathbf{v}_2$ by 2.

Cayley–Hamilton check: $A^2 - 7A + 10I = \begin{pmatrix} 18 & 7 \\ 14 & 11 \end{pmatrix} - \begin{pmatrix} 28 & 7 \\ 14 & 21 \end{pmatrix} + \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix} = 0$.

Not diagonalisable

$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$: $\lambda = 1$ twice (AM $= 2$), but $A - I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has nullity 1 (GM $= 1$). Only one independent eigenvector, so **not** diagonalisable.

Given eigenvalues $1, 2, 3$: $\det A = 6$, $\operatorname{tr} A = 6$; A^2 has $1, 4, 9$ ($\operatorname{tr} A^2 = 14$); A^{-1} has $1, \frac{1}{2}, \frac{1}{3}$; $A + 2I$ has $3, 4, 5$.

Common GATE traps

(1) Repeated eigenvalues do *not* automatically mean non-diagonalisable (e.g. I), but check GM vs AM. (2) Eigenvectors for the *same* eigenvalue are not necessarily orthogonal, only for distinct eigenvalues of a symmetric matrix. (3) tr and det give quick checks on your eigenvalue computation: always verify.

6 Quadratic Forms and Definiteness

Ramu & Somu: The shape of a bowl

Somu holds a quadratic function $q(x, y) = 2x^2 + 2xy + 2y^2$ and asks Ramu, "If I plot this, what shape do I get?" Ramu guesses a bowl: it is non-negative and zero only at the origin. Somu writes q with a **symmetric matrix**: $q = \mathbf{x}^\top A\mathbf{x}$ with $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. The matrix tells him the shape: eigenvalues 3 and 1 are both positive, so it is a bowl opening upward in every direction (**positive definite**). If one eigenvalue were negative, the surface would be a saddle.

6.1 Definitions

Formulas

A **quadratic form** in $\mathbf{x} \in \mathbb{R}^n$: $q(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x} = \sum_{i,j} a_{ij} x_i x_j$, with A taken **symmetric**. (A non-symmetric B gives the same form as $\frac{1}{2}(B + B^\top)$.)

To get A: diagonal entries = coefficients of x_i^2; off-diagonal $a_{ij} = a_{ji} = \frac{1}{2}\times$ (coefficient of $x_i x_j$).

Name	Condition on $\mathbf{x}^\top A\mathbf{x}$	Eigenvalues of symmetric A
Positive definite (PD)	> 0 for all $\mathbf{x} \neq 0$	all > 0
Positive semidefinite (PSD)	≥ 0 for all $\mathbf{x}$	all ≥ 0
Negative definite (ND)	< 0 for all $\mathbf{x} \neq 0$	all < 0
Negative semidefinite	≤ 0	all ≤ 0
Indefinite	takes both signs	both positive and negative

6.2 Tests for definiteness

Formulas

- **Eigenvalue test:** signs of the eigenvalues (above).
- **Sylvester's criterion:** A is PD $\iff$ all *leading principal minors* $\Delta_1, \Delta_2, \ldots, \Delta_n > 0$. A is ND $\iff$ signs alternate $\Delta_1 < 0, \Delta_2 > 0, \Delta_3 < 0, \ldots$
- **PSD needs *all* principal minors** ≥ 0 (not just the leading ones).
- 2×2: PD $\iff$ $a > 0$ and $ad - b^2 > 0$ (equivalently tr > 0 and det > 0).
- $B^\top B$ is always PSD; PD iff B has full column rank. Also A PD $\Rightarrow$ A invertible, $a_{ii} > 0$.
- **Diagonalising a form:** with $A = Q\Lambda Q^\top$ and $\mathbf{y} = Q^\top \mathbf{x}$: $q = \sum \lambda_i y_i^2$. **Rank** of the form = rank of A; **signature** = (#positive) − (#negative) eigenvalues. (Sylvester's law of inertia.)
- Link to calculus: the Hessian H at a critical point: PD $\Rightarrow$ local minimum, ND $\Rightarrow$ local maximum, indefinite $\Rightarrow$ saddle.
- **Rayleigh quotient:** $\lambda_{\min} \leq \dfrac{\mathbf{x}^\top A\mathbf{x}}{\mathbf{x}^\top \mathbf{x}} \leq \lambda_{\max}$ for symmetric A.

Worked example

$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$: leading minors $2,\ 3,\ 4$ are all positive, so A is **PD**. (Eigenvalues $2,\ 2\pm\sqrt{2}$, all positive; sum $= 6 = \operatorname{tr} A$, product $= 4 = \det A$.)

$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$: $\det = 1 - 4 = -3 < 0$, eigenvalues $3, -1$: **indefinite**.

$q = x^2 + 4xy + y^2$ has matrix $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ (the xy coefficient 4 is split as $2 + 2$): indefinite.

Common GATE traps

(1) Always symmetrise first. (2) Positive determinant alone does not imply PD: $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ has $\det = 1$ but is ND. (3) Sylvester's test for *semi*definiteness needs all principal minors, not just leading ones.

7 LU Decomposition

Ramu & Somu: Solving the same problem for many customers

Somu's stall has a fixed price matrix A but every day a different bill vector $\mathbf{b}$. Doing full elimination every day would be wasteful. Instead, Somu does the elimination *once*, records the steps in a matrix L and keeps the result in U. Now each new day costs only two cheap back-substitutions. That is LU decomposition: $A = LU$.

Formulas

- $A = LU$: L lower triangular with ones on the diagonal (stores the multipliers) , U upper triangular (the echelon form)
- Solving $A\mathbf{x} = \mathbf{b}$: $L\mathbf{y} = \mathbf{b}$ (forward substitution), $U\mathbf{x} = \mathbf{y}$ (back substitution)
- $\det A = \det U = \prod u_{ii}$ $(\det L = 1)$
- If a zero (or tiny) pivot appears, swap rows: $PA = LU$ (P = permutation matrix , $\det P = \pm 1$)
- Multiplier: $\ell_{ij} = \dfrac{a_{ij}}{a_{jj}}$ evaluated on the current (partly reduced) matrix. You do $R_i \leftarrow R_i - \ell_{ij} R_j$, and L stores ℓ_{ij}.
- Cost: elimination $\sim \frac{2}{3}n^3$ operations; each extra $\mathbf{b}$ costs only $\sim 2n^2$
- Symmetric positive definite A : Cholesky $A = LL^\top$ (half the cost), $A = LDL^\top$

A full LU example

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix}.$$

Step 1 (pivot 2): $R_2 \leftarrow R_2 - 2R_1$ $(\ell_{21} = 2)$, $R_3 \leftarrow R_3 - 4R_1$ $(\ell_{31} = 4)$: $\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 5 \end{pmatrix}$.

Step 2 (pivot 1): $R_3 \leftarrow R_3 - 3R_2$ $(\ell_{32} = 3)$: $U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$.

Hence

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}, \qquad LU = A.$$

Determinant: $\det A = 2 \cdot 1 \cdot 2 = 4$.
Solve $A\mathbf{x} = \mathbf{b}$ with $\mathbf{b} = (4, 10, 24)$: forward: $y_1 = 4$, $y_2 = 10 - 2(4) = 2$, $y_3 = 24 - 4(4) - 3(2) = 2$.
Back: $x_3 = 2/2 = 1$, $x_2 = 2 - 1 = 1$, $x_1 = (4 - 1 - 1)/2 = 1$. So $\mathbf{x} = (1, 1, 1)$. ✓

Common GATE traps

(1) The multipliers go into L with a *positive* sign (you subtract $\ell_{ij}\times$row, so L stores $+\ell_{ij}$). (2) Without row swaps LU exists only if all leading principal minors are nonzero. (3) $\det A = \prod u_{ii}$ only when no row swaps happened; with swaps multiply by $(-1)^{\#\text{swaps}}$.

8 Singular Value Decomposition (SVD)

Ramu & Somu: Every matrix is a rotate–stretch–rotate

Ramu: "Eigen-decomposition only works for square, nice matrices. What about a 2×3 table of ratings?" Somu: SVD works for *every* matrix. Any linear map can be seen as three simple steps: (1) rotate/reflect the input space ($V^\top$), (2) stretch along the coordinate axes by non-negative numbers σ_i (Σ), (3) rotate/reflect in the output space (U). The stretch factors σ_i are the **singular values**: how strongly the matrix acts in each of its principal directions.

Formulas

- $A = U\Sigma V^\top, \qquad A \in \mathbb{R}^{m\times n}, \quad U \in \mathbb{R}^{m\times m}$ orthogonal , $V \in \mathbb{R}^{n\times n}$ orthogonal
- $\Sigma \in \mathbb{R}^{m\times n}$ is "diagonal" with $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ and zeros elsewhere , $\quad r = \operatorname{rank} A$
- $\sigma_i = \sqrt{\lambda_i(A^\top A)} = \sqrt{\lambda_i(AA^\top)}$ (non-zero eigenvalues of $A^\top A$ and $AA^\top$ coincide)
- V : eigenvectors of $A^\top A$ (right singular vectors), $\quad U$: eigenvectors of $AA^\top$ (left singular vectors)
- $A\mathbf{v}_i = \sigma_i \mathbf{u}_i \;\Rightarrow\; \mathbf{u}_i = \frac{A\mathbf{v}_i}{\sigma_i} \qquad (i = 1, \ldots, r)$
- Outer-product form: $A = \sum_{i=1}^{r} \sigma_i \, \mathbf{u}_i \mathbf{v}_i^\top$

8.1 What SVD tells you

Formulas

- $\operatorname{rank} A = \#\{\sigma_i > 0\}$
- $\|A\|_2 = \sigma_1$ (largest stretch), $\qquad \|A\|_F = \sqrt{\sum \sigma_i^2} = \sqrt{\operatorname{tr}(A^\top A)}$
- Condition number: $\kappa(A) = \sigma_{\max}/\sigma_{\min}$ (square invertible A)
- $|\det A| = \prod \sigma_i$ (square A)
- Column space $\mathcal{C}(A) = \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}; \quad \mathcal{N}(A) = \operatorname{span}\{\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n\}$
- **Eckart–Young:** best rank- k approximation: $A_k = \sum_{i=1}^{k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top, \quad \|A - A_k\|_2 = \sigma_{k+1}, \quad \|A - A_k\|_F^2 = \sum_{i>k} \sigma_i^2$
- Pseudo-inverse: $A^+ = V\Sigma^+ U^\top$ (invert non-zero σ_i) least-squares solution $\hat{\mathbf{x}} = A^+\mathbf{b}$
- Symmetric PSD A : singular values = eigenvalues $\qquad$ symmetric A : $\sigma_i = |\lambda_i|$
- PCA: for centred data X, principal directions = right singular vectors of X; variance along $\mathbf{v}_i = \sigma_i^2/(n-1)$

A full SVD, step by step

Let $A = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}$ (2×3).

Step 1. $AA^\top = \begin{pmatrix} 17 & 8 \\ 8 & 17 \end{pmatrix}$ (check: $9+4+4=17$, $6+6-4=8$). Eigenvalues $17 \pm 8 = 25,\ 9$. So **singular values** $\sigma_1 = 5,\ \sigma_2 = 3$, rank $A = 2$.
Step 2. Eigenvectors of $AA^\top$: $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1)$ for 25, $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)$ for 9.
Step 3. $\mathbf{v}_i = A^\top \mathbf{u}_i/\sigma_i$: $A^\top(1,1) = (5,5,0)$, so $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1,0)$; $A^\top(1,-1) = (1,-1,4)$, so $\mathbf{v}_2 = \frac{1}{3\sqrt{2}}(1,-1,4)$.
Step 4. A third orthonormal $\mathbf{v}_3 \in \mathcal{N}(A)$: $\mathbf{v}_3 = \frac{1}{3}(-2,2,1)$. Check $A\mathbf{v}_3 = (-6+4+2,\ -4+6-2) = (0,0)$. ✓
Therefore $A = U\Sigma V^\top$ with $\Sigma = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$. Here $\|A\|_2 = 5$, $\|A\|_F = \sqrt{25+9} = \sqrt{34}$, and the best rank-1 approximation is $5\,\mathbf{u}_1\mathbf{v}_1^\top = \frac{5}{2}\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$.

Ramu & Somu: Compressing a photo

Ramu stores a 1000×1000 photo as a matrix: a million numbers. Somu does SVD and notices the singular values drop quickly: $\sigma_1, \ldots, \sigma_{50}$ are large and the rest tiny. Keeping only the top 50 terms stores $50 \times (1000 + 1000 + 1) \approx 100{,}000$ numbers (a $10\times$ saving) with very little visible loss. By Eckart–Young no other rank-50 matrix is closer to the original. This is how SVD powers image compression, recommender systems, PCA and noise removal.

Common GATE traps

(1) **Singular values are not eigenvalues** unless A is symmetric PSD. (2) $\sigma_i \geq 0$ always, and they are sorted in decreasing order. (3) The *number* of non-zero singular values equals the rank, and Σ has the same shape as A ($m \times n$). (4) Sign choices in $\mathbf{u}_i, \mathbf{v}_i$ are not unique but must be consistent: $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$.

9 One-page Formula Sheet

Topic	Remember
Independence	columns of A independent $\iff$ $\operatorname{rank} A$ = number of columns $\iff$ $\mathcal{N}(A) = \{0\}$
Dimension	$\dim \mathcal{C} = \dim \text{Row} = r$; $\dim \mathcal{N}(A) = n - r$; $\dim \mathcal{N}(A^\top) = m - r$
Transpose / inverse	$(AB)^\top = B^\top A^\top$; $(AB)^{-1} = B^{-1}A^{-1}$
Determinant	$\det(AB) = \det A \det B$; $\det(cA) = c^n \det A$; $\det = \prod \lambda_i$
Systems	$r < r_{aug}$: none; $r = r_{aug} = n$: unique; $r = r_{aug} < n$: infinite ($n - r$ free)
Homogeneous	non-trivial solution $\iff$ $r < n$ ($\iff$ $\det A = 0$ if square)
Rank–Nullity	$\operatorname{rank} A + \operatorname{nullity} A = n$
Eigenvalues	$\det(A - \lambda I) = 0$; $\sum \lambda = \text{tr}$; $\prod \lambda = \det$; $A^k \to \lambda^k$; $A^{-1} \to 1/\lambda$
Diagonalisation	GM = AM for all λ; $A = PDP^{-1}$; symmetric: $A = Q\Lambda Q^\top$
Quadratic forms	PD $\iff$ all $\lambda > 0$ $\iff$ all leading minors > 0
LU	$A = LU$; $\det A = \prod u_{ii}$; solve $L\mathbf{y} = \mathbf{b}$ then $U\mathbf{x} = \mathbf{y}$
SVD	$A = U\Sigma V^\top$; $\sigma_i = \sqrt{\lambda_i(A^\top A)}$; rank $= \#\sigma_i > 0$; $\|A\|_2 = \sigma_1$; $\|A\|_F^2 = \sum \sigma_i^2$
Least squares	$A^\top A\hat{\mathbf{x}} = A^\top \mathbf{b}$; projection $P = A(A^\top A)^{-1}A^\top$

10 Practice Questions (with answers)

Try each on your own first.

1. Rank of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$? **Ans:** 1 (all rows are multiples of row 1).

2. A is 4×6 with rank $= 3$. Find nullity(A) and $\dim \mathcal{N}(A^\top)$. **Ans:** 3 and 1.

3. A has eigenvalues $1, 2, 3$. Find $\det A$, $\text{tr}\, A^2$, eigenvalues of A^{-1}. **Ans:** 6; 14; $1, \frac{1}{2}, \frac{1}{3}$.

4. $x + y = 2$, $x + y = 3$: how many solutions? **Ans:** none ($\operatorname{rank} A = 1 < \operatorname{rank}[A|b] = 2$).

5. $\det \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $\det(2A)$? **Ans:** -2 and -8.

6. Is $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ positive definite? **Ans:** No; eigenvalues $3, -1$: indefinite.

7. $A^\top A$ has eigenvalues 16 and 4. Find $\|A\|_2$ and $\|A\|_F$. **Ans:** 4 and $\sqrt{20}$.

8. The singular values of A are $5, 3, 0$. Rank of A? **Ans:** 2.

9. 3 equations, 5 unknowns, $\operatorname{rank} A = \operatorname{rank}[A|b] = 3$. Free variables? **Ans:** 2 (infinitely many solutions).

10. Is $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ diagonalisable? **Ans:** No (AM = 2, GM = 1).

11. 3×3 A has eigenvalues $2, 3, -1$. Find $\det(A^\top A)$. **Ans:** $(\det A)^2 = (-6)^2 = 36$.

12. From $A = LU$ with $U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ and no row swaps, find $\det A$. **Ans:** $2 \cdot 1 \cdot 2 = 4$.

13. Matrix of $q = x^2 + 4xy + y^2$? **Ans:** $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, indefinite.

14. $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$: eigenvalues? **Ans:** 5 and 2.

How to study this section: go in order: vector spaces → matrix types and determinant → systems → rank–nullity → eigenvalues → quadratic forms → LU → SVD. For each topic derive the key formula once, then solve 15–20 GATE-style questions. Highest-yield areas for GATE DA: rank and nullity with systems, eigenvalue properties (trace, determinant, powers), definiteness tests, and the link between SVD and eigenvalues of $A^\top A$. Linear algebra is also the base of PCA, regression and neural networks in the Machine Learning section.